

MOSTOFA HASIN MAHDI

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SUMMARY

Final-year Computer Science and Engineering student with hands-on experience building backend APIs, data-driven applications, and AI/ML systems using Python, FastAPI, React, PostgreSQL, and Scikit-learn. Experienced in developing OCR pipelines, multi-tenant systems, API validation workflows, and machine-learning applications.

EDUCATION

Shanto-Mariam University of Creative Technology <i>B.Sc. in Computer Science and Engineering</i> CGPA: 3.96	Dhaka, Bangladesh 2022 – Expected Nov 2026
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PROJECTS

NAB Preg AI <i>Lead ML Engineer (Team of 5)</i>	<i>FastAPI, Next.js, XGBoost, Scikit-learn, Mistral AI, Supabase</i>
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- Built a multi-output ML model predicting 3 clinical risk targets in **<50ms** with **90% accuracy**; mitigated class imbalance by augmenting **740 records with 400 synthetic high-risk profiles** using WHO-aligned thresholds.
- Developed an **OCR pipeline** integrating Mistral's **Pixtral-12B** via FastAPI, parsing prescriptions and lab reports into sanitized JSON for frontend validation.
- Engineered a **5-step OCR failover** pipeline using 4 backup Mistral API keys and Tesseract.js as a local fallback.
- Implemented dual-layer Zod/Pydantic validation to sanitize 10 patient fields and prevent malformed database payloads.
- *Demo:* [Live Demo](#) · [GitHub](#)

PDF to Markdown Conversion Engine <i>Solo Developer</i>	<i>PyMuPDF, pdfplumber, Pandas, spaCy, PyTesseract</i>
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- Developed an open-source Python CLI using custom **coordinate-based** layout detection to sort text, detect headings, and prevent overlapping content.
- Integrated **spaCy-based text cleaning** to resolve sentence fragmentation and PyTesseract OCR for scanned pages.
- Engineered table extraction that identifies complex grids and reconstructs them into aligned Markdown tables.
- *Demo:* [GitHub](#) · [Release](#)

Bizit Business Suite <i>Solo Developer</i>	<i>React, FastAPI, Pydantic, PostgreSQL, Pytest</i>
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- Engineered a secure multi-tenant architecture using PostgreSQL data scoping, ensuring **zero cross-tenant leaks**.
- Implemented 3-tier RBAC (Owner/Admin/Employee) and Pydantic validation across **24 API endpoints**.
- Optimized PostgreSQL queries to aggregate **10,000+ sales and cost records** with **sub-10ms** response times.
- Wrote **17 Pytest integration tests** covering Stock, Sales, Suppliers, RBAC, and tenant isolation, executed automatically on every push via GitHub Actions CI.
- Containerized frontend and backend services with **Docker**, using local PostgreSQL to mirror production architecture.
- *Demo:* username: hasin, password: hasin123 · [Live Demo](#) · [GitHub](#)

TECHNICAL SKILLS

Languages:	Python, C++, JavaScript, TypeScript, SQL
Backend/Frontend:	FastAPI, Next.js, React, REST APIs, Pydantic
AI/ML:	Scikit-learn, XGBoost, Pandas, OCR, NLP
Databases:	PostgreSQL, MongoDB, MySQL
DevOps & Tools:	Docker, Git, GitHub Actions, Pytest, Linux

ACHIEVEMENTS

- **Champion**, SMUCT Software Project Showcase 2026; **Finalist**, CloudcampBD Infinity AI BuildFest 2026.
- **2nd Runner-up**, SMUCT CSE FEST V2.0 Programming Contest.